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MEDICINE

Shoulder Dystocia

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Objectives

- 1-Identify the risk factors for shoulder dystocia (SD)
- 2-Describe a systematic team based approach to managing SD
- 3-Explain the appropriate maneuvers to reduce SD using the HELPER[4] mnemonic

Diagnosis of Shoulder Dystocia

- Definition:
- Shoulder dystocia is defined as a vaginal cephalic delivery that requires additional obstetric manoeuvres to deliver the fetus after the head has delivered and gentle traction has been unsuccessful in delivering the shoulders .
  - Head-to-body delivery time interval of ≥60 seconds
  - Use of any ancillary obstetric maneuvers to affect delivery

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Incidence

- 0.1% to 3% of all deliveries
- Prediction of macrosomia by ultrasonography is modest at best
- More than 50% of all SDs occur in normal birth weight infants

Shoulder dystocia: prevention/risk factors/warning signs

- Prevention: diagnosis and optimal control of gestational and insulin-dependent diabetics, reduction of maternal obesity.
- Careful plan for mode of delivery in women with previous shoulder dystocia delivery (recurrence rate 10–15%).
- Risk factors:
- macrosomia, poorly controlled gestational and insulin-dependent diabetes, maternal obesity, previous shoulder dystocia, instrumental,prolonged first stage or second stage of labour.

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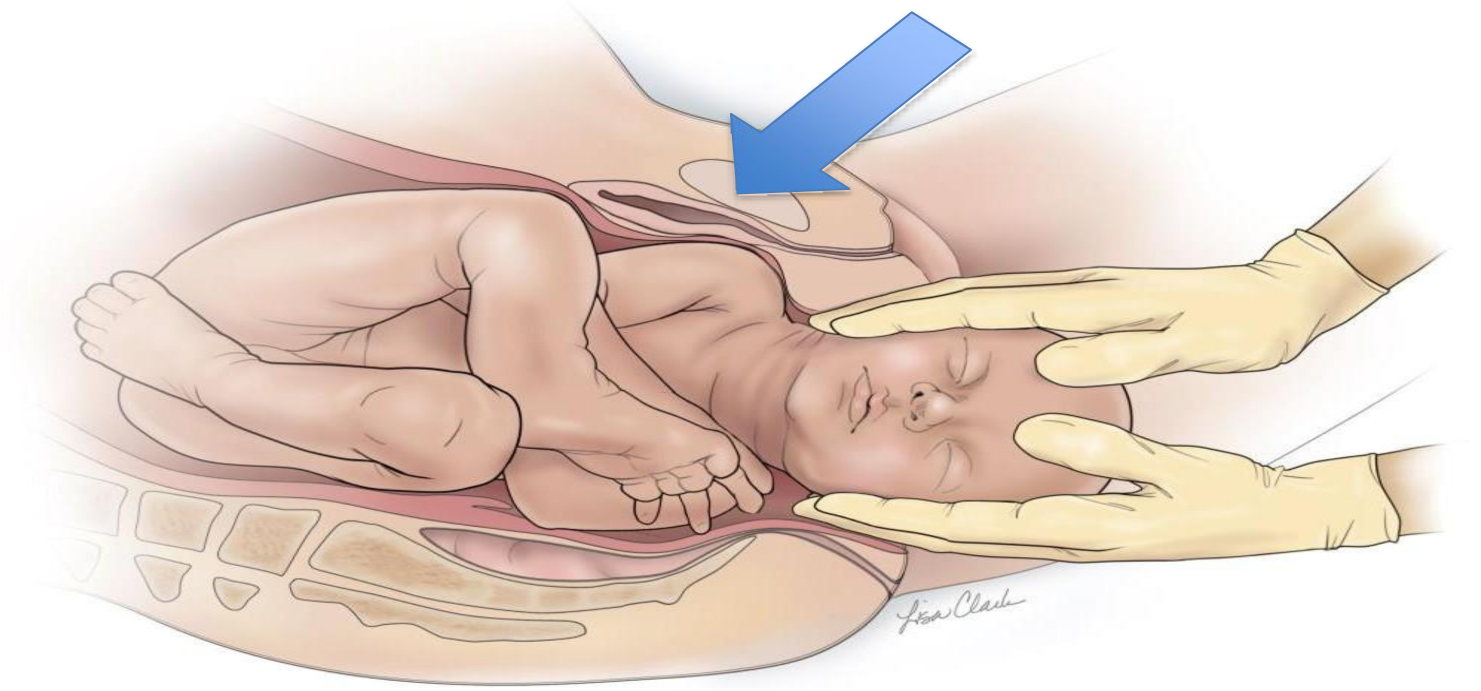
Turtule sign

- Warning signs:
- failure of restitution of head following delivery of the head, retraction of the fetal head against the perineum (analogous to a turtle withdrawing into its shell)

Impacted Shoulder



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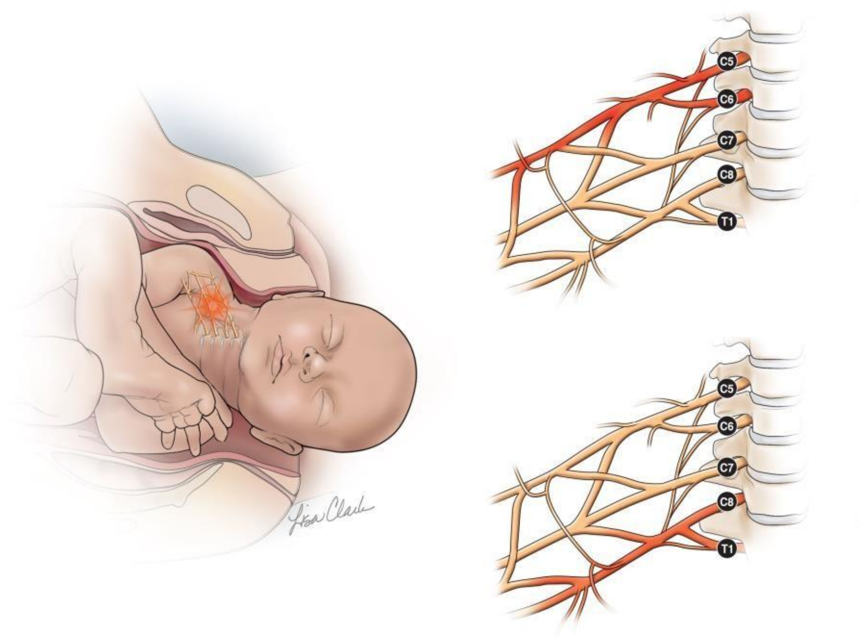
Brachial Plexus Palsies



- Erb Palsy
  - C5-C6 nerve root
  - Loss of sensation in arm
  - Paralysis of deltoid, biceps, and brachialis muscles
- Klumpke Palsy
  - C5-C6 nerve root
  - C8-T1 nerve root

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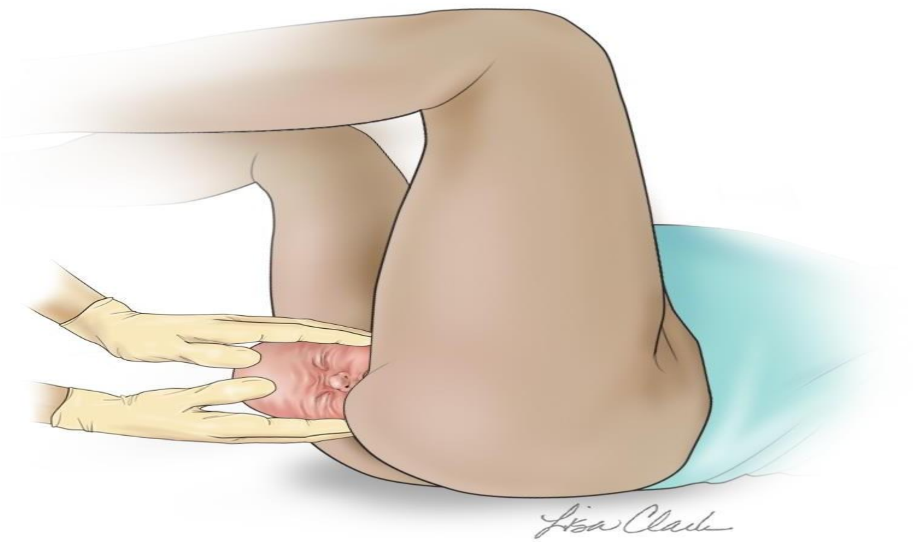
Recognition of Shoulder Dystocia

Fetal head retracts against the perineum (*turtle sign*)
Gentle axial traction does not affect delivery
Proceed to the

HELPER[4] mnemonic

Example of *Turtle Sign*



What is the Safe Amount of Time to Deliver?

- The safe amount of time to avoid fetal acidosis caused by SD is unknown
  - Neonatal pH level <7 is associated with asphyxia and difficult resuscitation
  - Increased risk of acidosis if >5 minutes have elapsed
  - Each intervention takes 30 to 60 seconds and provides a safe window of time to accomplish vaginal delivery

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- **Mnemonic for Shoulder Dystocia**
- **Management**
- HELPER[4]

- **Managing Shoulder Dystocia is a Team Effort**

H = Call for *H*elp Early

- After recognition of SD, activate institutional protocol
  - Appropriate notification
    - State out loud, “We have a shoulder dystocia.”
  - Additional nursing staff
  - Additional backup
    - Extra nursing staff
    - Neonatal resuscitation team
    - Obstetric/surgical staff
    - Anesthesia staff

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E = *E*valuate and *E*xplain

- Shoulder dystocia is not a soft-tissue dystocia

- Explain the situation to patient

- Explain to team members that a SD is present, which shoulder is anterior, and assign roles
    - Episiotomy

L = Legs

Flex maternal hips at a 90-degree angle (not out to the side) so that thighs are just touching the sides of abdomen

- Effect
  - Straightens the lumbosacral lordosis
  - Increases the anteroposterior diameter of pelvis
  - Flexes the fetal spine
  - **Resolves 42% of SDs**

- McRoberts Maneuver

P = *P*ressure

McRoberts Maneuver Combine With Suprapubic Pressure

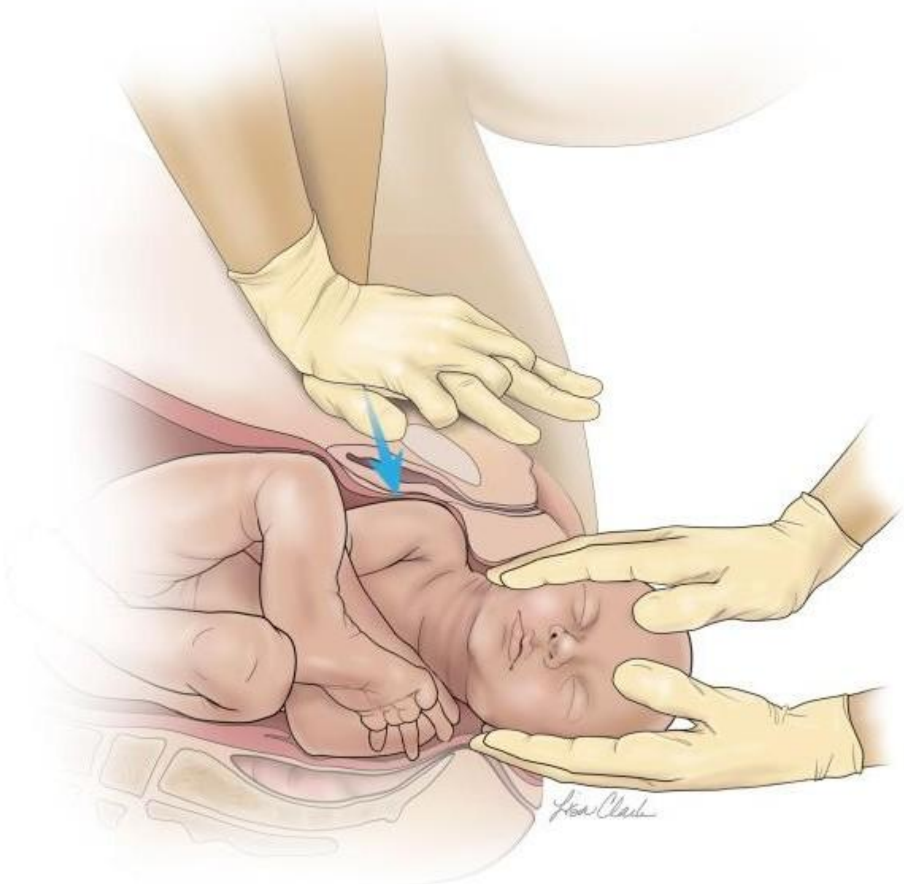
Suprapubic pressure performed by assistant

- Locking hand position similar to cardiopulmonary resuscitation
- Force should act to adduct the anterior shoulder
- initially apply continuous pressure,

but can tlhen involve a rocking motion

- Attempt for 30 to 60 seconds

No fundal pressure



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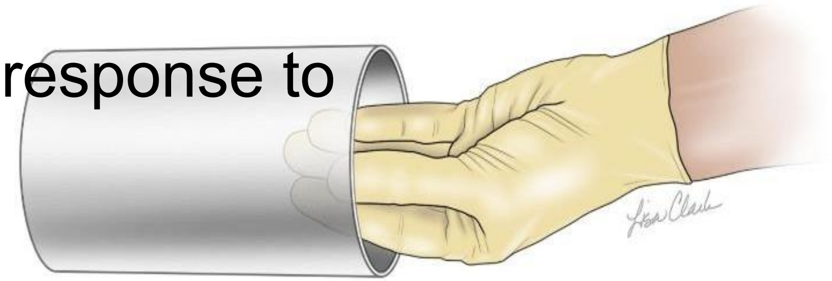
E = *E*nter +/- Episiotomy

- Consider when additional room for the

clinician’s hand is needed to perform internal maneuvers

- Decision based on clinical judgment and response to initial maneuvers



Techniques for Delivering the Posterior Arm

- Confirm position of fetus (aim for front of chest)
- Make entering hand small
- Introduce appropriate hand into introitus at 6 o'clock position
- Follow fetus’s chest to find forearm
- If not found, arm will be behind back change hands!

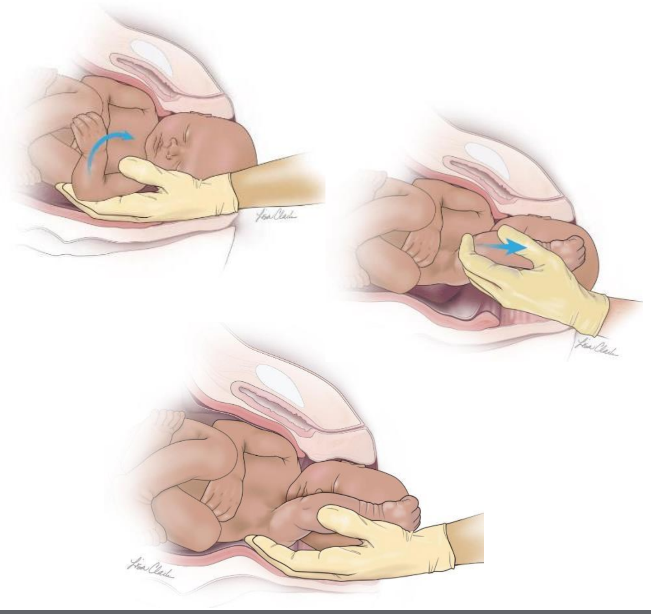
R^4 = Remove, Rotate, Roll, Repeat

• *R^4 = Remove, Rotate, Roll, Repeat*

- **R**emoval of the posterior arm
- Use of **R**otatory internal maneuvers (previously known as the Enter maneuvers):
  - Rubin II
  - Rubin II + Woods screw
  - Reverse Woods screw
- **R**oll the patient (Gaskin maneuver)
- **R**epeat maneuvers as necessary

R^1= *R*emoval of the Posterior Arm

- Delivery of posterior arm is the safest and most effective maneuver



AAFP Advanced Life Support in Obstetrics

  - Follow the posterior arm down to the elbow, usually located anterior to the fetal chest
- Flexw the arm at the elbow
- Seep the forearm across the fetal chest
  - – Grasping the humerus or hand should be avoided because it can lead to fractures
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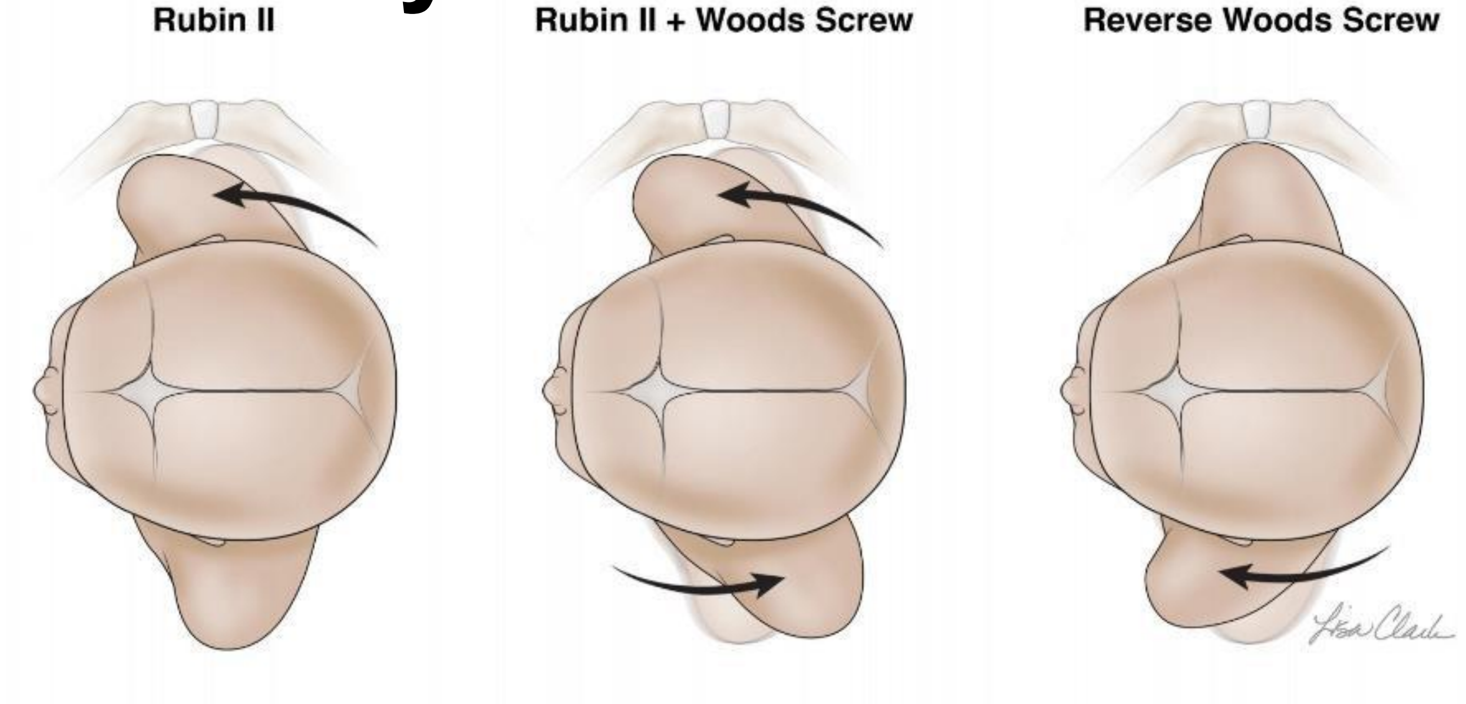
R^2 = *R*otatory Internal Maneuvers



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•Rubin II Maneuver

- Approach anterior fetal shoulder by inserting index and middle finger into introitus at 5 o'clock (if left occiput
  - transverse) or 7 o’clock (if right occiput transverse) position
- Sweep fingers into a position behind the anterior shoulder
- Exert pressure to adduct most accessible shoulder and rotate to oblique position
- Continue McRoberts maneuver

•Rubin II + Woods Screw Maneuver

- Keep fingers that are behind the anterior shoulder in place
- Place two fingers from the opposite hand in front of the posterior fetal shoulder
- Discontinue suprapubic pressure
- Gently rotate shoulders
  - – Clinician has one hand on each shoulder, rotating together
-

•Reverse Woods Screw

- Remove fingers that are in front of the posterior shoulder from the vagina
- Sweep fingers that were behind the anterior shoulder to a position behind the posterior shoulder
- Rotate fetus in the opposite direction from Woods screw maneuver

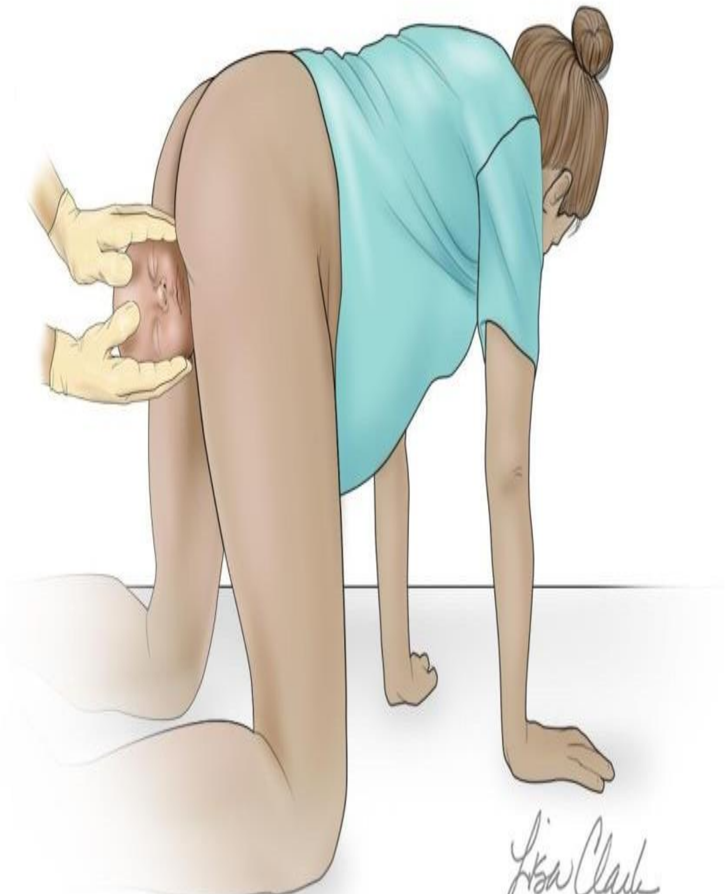
R^3 = *R*oll the Patient

•Reposition into all-fours

Increases pelvic diameters

•Movement and gravity may also contribute to dislodging the impacted shoulder

•Deliver the shoulder that is located closest to the ceiling with gentle axial traction



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•R^4 = *Repeat*

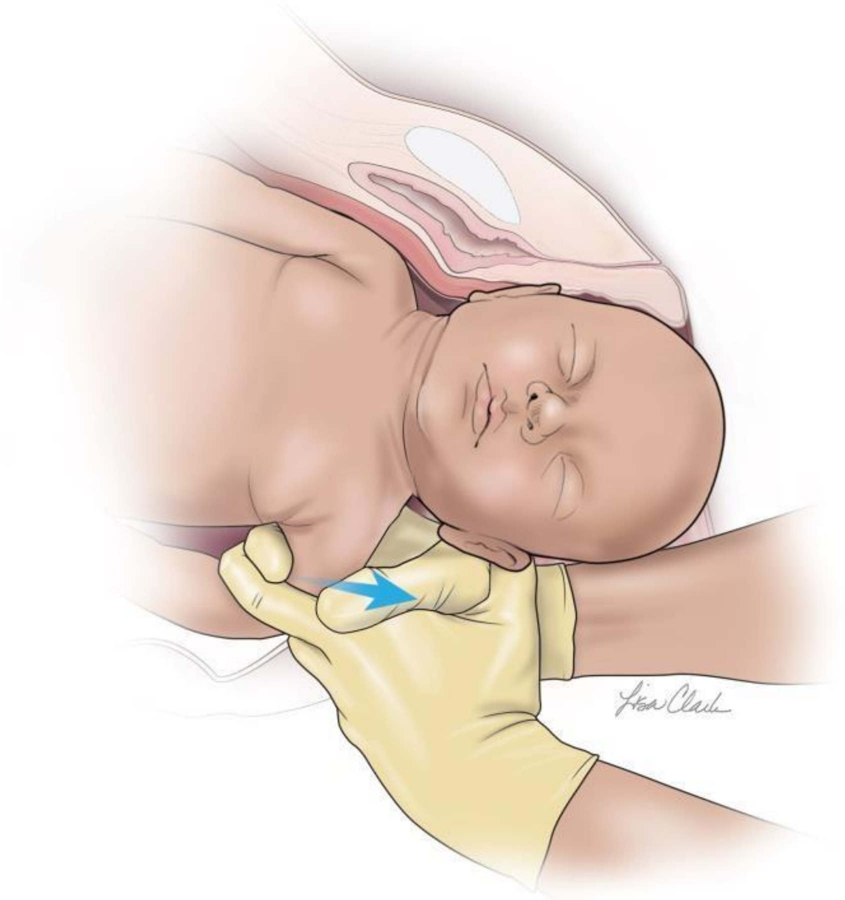
•

- Repeat all maneuvers in all-fours position or in the lithotomy position if the woman has not been moved to the all-fours position

•Additional Maneuvers

- Posterior axillary sling traction
- The methods of last resort
  - Zavanelli maneuver
  - Symphysiotomy
  - Abdominal rescue (cesarean delivery)

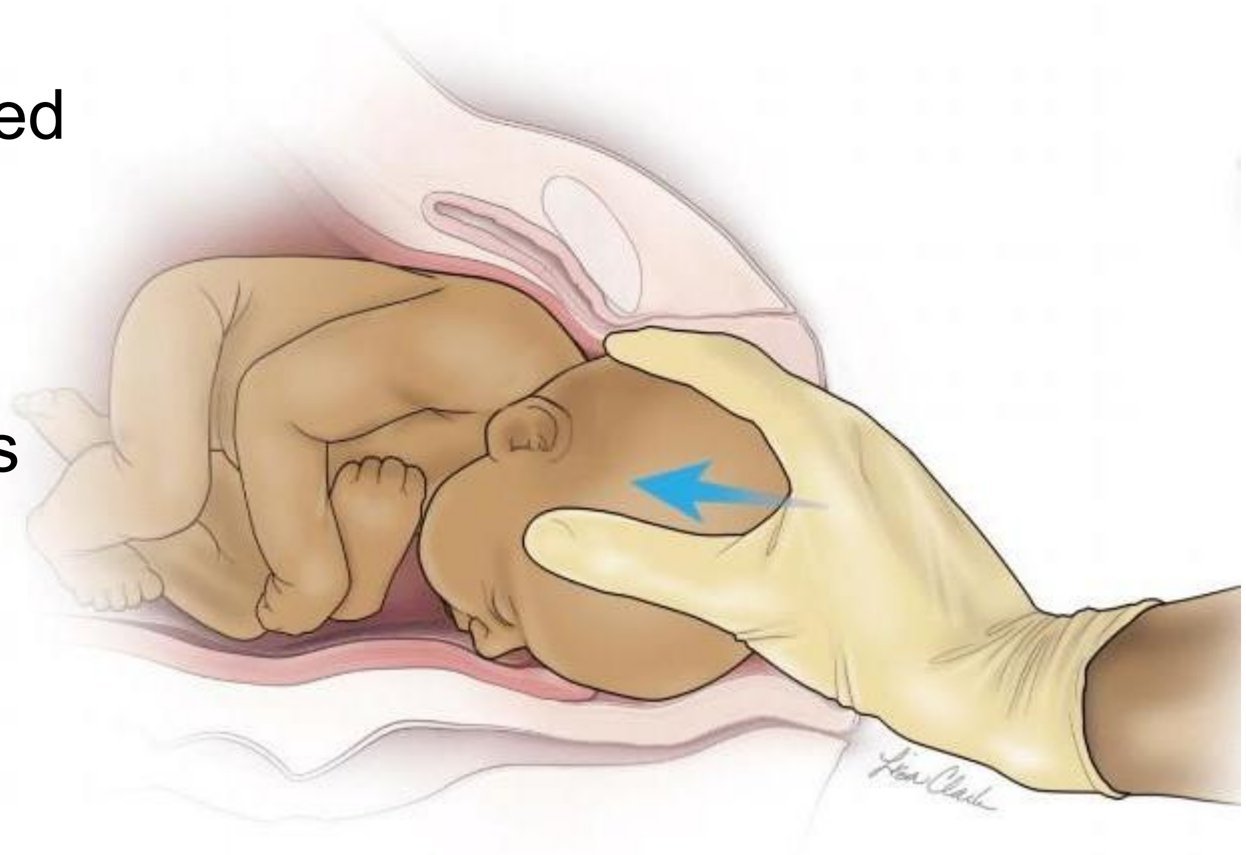
Posterior Axillary Sling Traction



Zavanelli Maneuver

Flex fetal head to replace

- Cephalic replacement followed by emergency cesarean delivery
- Requires anesthesia, operative team, and tocolysis
- Not an option if nuchal cord has been clamped and cut



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Complications of Shoulder Dystocia

- Maternal
  - Soft-tissue injuries
  - Anal sphincter damage
  - Postpartum hemorrhage
  - Uterine rupture
  - Symphyseal separation
  - Transient femoral neuropathy
- Neonatal
  - Brachial plexus palsy
  - Clavicle fracture
  - Humeral fracture
  - Fetal acidosis
  - Hypoxic brain injury
  - Neonatal death

Documentation of Shoulder Dystocia

- Procedure note should include:
  - Elapsed time of the SD
  - Which maneuvers were attempted and in what order
  - Which shoulder/arm was anterior
  - Fetal head position
  - Assessment of the woman and infant after delivery
  - Team members present
- Umbilical cord pH results